LAB ASSIGNMENT – 7 (Studio 3t)
DATE – 23.03.2026

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REGISTRATION NO. : AU/2023/0009045

ROLL NO. : UG/02/BTCSECSF/2023/003

SUBJECT : Unstructured Database Lab
SEC: E

Data: { roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },

{ roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78 },

{ roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 },

{ roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74 },

{ roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 },

{ roll: 6, name: "Simran", age: 22, department: "IT", marks: 81 },

{ roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69 },

{ roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95 }.

Lab Questions:

Q1. Display all documents.

```
test> use studentDB
switched to db studentDB
studentDB> db.students.insertMany([
... { roll: 1, name: "Anu", age: 22, department: "IT", marks: 85 },
... { roll: 2, name: "Riya", age: 21, department: "CSE", marks: 78 },
... { roll: 3, name: "Aman", age: 23, department: "IT", marks: 92 },
... { roll: 4, name: "Neha", age: 20, department: "ECE", marks: 74 },
... { roll: 5, name: "Raj", age: 24, department: "CSE", marks: 88 },
... { roll: 6, name: "Simran", age: 22, department: "IT", marks: 81 },
... { roll: 7, name: "Karan", age: 21, department: "ECE", marks: 69 },
... { roll: 8, name: "Priya", age: 23, department: "CSE", marks: 95 }
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69c0e37495575397b61e2621'),
    '1': ObjectId('69c0e37495575397b61e2622'),
    '2': ObjectId('69c0e37495575397b61e2623'),
    '3': ObjectId('69c0e37495575397b61e2624'),
    '4': ObjectId('69c0e37495575397b61e2625'),
    '5': ObjectId('69c0e37495575397b61e2626'),
    '6': ObjectId('69c0e37495575397b61e2627'),
    '7': ObjectId('69c0e37495575397b61e2628')
  }
}
```

Q2. Find students from IT department.

```
studentDB> db.students.find()
[
  {
    _id: ObjectId('69c0e37495575397b61e2621'),
    roll: 1,
    name: 'Anu',
    age: 22,
    department: 'IT',
    marks: 85
  },
  {
    _id: ObjectId('69c0e37495575397b61e2622'),
    roll: 2,
    name: 'Riya',
    age: 21,
    department: 'CSE',
    marks: 78
  },
  {
    _id: ObjectId('69c0e37495575397b61e2623'),
    roll: 3,
    name: 'Aman',
    age: 23,
    department: 'IT',
    marks: 92
  },
  {
    _id: ObjectId('69c0e37495575397b61e2624'),
    roll: 4,
    name: 'Neha',
    age: 20,
    department: 'ECE',
    marks: 74
  },
  {
    _id: ObjectId('69c0e37495575397b61e2625'),
    roll: 5,
    name: 'Raj',
    age: 24,
    department: 'CSE',
    marks: 88
  },
```

```
  {
    _id: ObjectId('69c0e37495575397b61e2626'),
    roll: 6,
    name: 'Simran',
    age: 22,
    department: 'IT',
    marks: 81
  },
  {
    _id: ObjectId('69c0e37495575397b61e2627'),
    roll: 7,
    name: 'Karan',
    age: 21,
    department: 'ECE',
    marks: 69
  },
  {
    _id: ObjectId('69c0e37495575397b61e2628'),
    roll: 8,
    name: 'Priya',
    age: 23,
    department: 'CSE',
    marks: 95
  }
]
studentDB> db.students.find({ department: "IT" }
[
  {
    _id: ObjectId('69c0e37495575397b61e2621'),
    roll: 1,
    name: 'Anu',
    age: 22,
    department: 'IT',
    marks: 85
  },
  {
    _id: ObjectId('69c0e37495575397b61e2623'),
    roll: 3,
    name: 'Aman',
    age: 23,
    department: 'IT',
    marks: 92
  },
  {
    _id: ObjectId('69c0e37495575397b61e2626'),
    roll: 6,
    name: 'Simran',
    age: 22,
    department: 'IT',
    marks: 81
  }
]
```

Q3. Find students with marks > 80.

```
studentDB> db.students.find({ marks: { $gt: 80 } })
[
  {
    _id: ObjectId('69c0e37495575397b61e2621'),
    roll: 1,
    name: 'Anu',
    age: 22,
    department: 'IT',
    marks: 85
  },
  {
    _id: ObjectId('69c0e37495575397b61e2623'),
    roll: 3,
    name: 'Aman',
    age: 23,
    department: 'IT',
    marks: 92
  },
  {
    _id: ObjectId('69c0e37495575397b61e2625'),
    roll: 5,
    name: 'Raj',
    age: 24,
    department: 'CSE',
    marks: 88
  },
  {
    _id: ObjectId('69c0e37495575397b61e2626'),
    roll: 6,
    name: 'Simran',
    age: 22,
    department: 'IT',
    marks: 81
  },
  {
    _id: ObjectId('69c0e37495575397b61e2628'),
    roll: 8,
    name: 'Priya',
    age: 23,
    department: 'CSE',
    marks: 95
  }
]
```

Q4. Insert one new student.

```
studentDB> db.students.insertOne({ roll: 9, name: "Arjun", age: 22, departmedepartment: "IT", marks: 87 })
{
  acknowledged: true,
  insertedId: ObjectId('69c0e38f95575397b61e2629')
}
```

Q5. Update marks of Anu to 90.

```
studentDB> db.students.updateOne(
...   { name: "Anu" },
...   { $set: { marks: 90 } }
... )
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

Q6. Add a new field "status: Pass" for all students.

```
studentDB> db.students.updateMany({}, { $set: { status: "Pass" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 9,
  modifiedCount: 9,
  upsertedCount: 0
}
```

Q7. Delete all students with marks < 70.

```
studentDB> db.students.deleteMany({ marks: { $lt: 70 } })
{ acknowledged: true, deletedCount: 1 }
```

Q8. Count students in CSE department.

```
studentDB> db.students.countDocuments({ department: "CSE" })
3
```

Q9. Find total students in each department.

```
studentDB> db.students.aggregate([
...   { $group: { _id: "$department", total: { $sum: 1 } } }
... ])
[
  { _id: 'CSE', total: 3 },
  { _id: 'IT', total: 4 },
  { _id: 'ECE', total: 1 }
]
```

Q10. Find highest marks.

```
studentDB> db.students.find().sort({ marks: -1 }).limit(1)
[
  {
    _id: ObjectId('69c0e37495575397b61e2628'),
    roll: 8,
    name: 'Priya',
    age: 23,
    department: 'CSE',
    marks: 95,
    status: 'Pass'
  }
]
```

Q11. Find the second highest marks.

```
studentDB> db.students.find().sort({ marks: -1 }).skip(1).limit(1)
[
  {
    _id: ObjectId('69c0e37495575397b61e2623'),
    roll: 3,
    name: 'Aman',
    age: 23,
    department: 'IT',
    marks: 92,
    status: 'Pass'
  }
]
```

Q12. Find students whose name starts with 'A' and marks > 80.

```
studentDB> db.students.find({
...   name: { $regex: "^A" },
...   marks: { $gt: 80 }
... })
[
  {
    _id: ObjectId('69c0e37495575397b61e2621'),
    roll: 1,
    name: 'Anu',
    age: 22,
    department: 'IT',
    marks: 90,
    status: 'Pass'
  },
  {
    _id: ObjectId('69c0e37495575397b61e2623'),
    roll: 3,
    name: 'Aman',
    age: 23,
    department: 'IT',
    marks: 92,
    status: 'Pass'
  },
  {
    _id: ObjectId('69c0e38f95575397b61e2629'),
    roll: 9,
    name: 'Arjun',
    age: 22,
    department: 'IT',
    marks: 87,
    status: 'Pass'
  }
]
```

Q13. Display department-wise average marks, only for departments where avg > 80.

```
studentDB> db.students.aggregate([
...   { $group: { _id: "$department", avgMarks: { $avg: "$marks" } } },
...   { $match: { avgMarks: { $gt: 80 } } }
... ])
[ { _id: 'IT', avgMarks: 87.5 }, { _id: 'CSE', avgMarks: 87 } ]
```

Q14. Find students with marks between 75 and 90, sorted by marks ascending.

```
studentDB> db.students.find({
...   marks: { $gte: 75, $lte: 90 }
... }).sort({ marks: 1 })
[
  {
    _id: ObjectId('69c0e37495575397b61e2622'),
    roll: 2,
    name: 'Riya',
    age: 21,
    department: 'CSE',
    marks: 78,
    status: 'Pass'
  },
  {
    _id: ObjectId('69c0e37495575397b61e2626'),
    roll: 6,
    name: 'Simran',
    age: 22,
    department: 'IT',
    marks: 81,
    status: 'Pass'
  },
  {
    _id: ObjectId('69c0e38f95575397b61e2629'),
    roll: 9,
    name: 'Arjun',
    age: 22,
    department: 'IT',
    marks: 87,
    status: 'Pass'
  },
  {
    _id: ObjectId('69c0e37495575397b61e2625'),
    roll: 5,
    name: 'Raj',
    age: 24,
    department: 'CSE',
    marks: 88,
    status: 'Pass'
  },
  {
    _id: ObjectId('69c0e37495575397b61e2621'),
    roll: 1,
    name: 'Anu',
    age: 22,
    department: 'IT',
    marks: 90,
    status: 'Pass'
  }
]
```

Q15. Add grade field based on marks.

```
studentDB> db.students.updateMany({}, [
...   {
...     $set: {
...       grade: {
...         $switch: {
...           branches: [
...             { case: { $gte: ["$marks", 90] }, then: "A" },
...             { case: { $gte: ["$marks", 80] }, then: "B" },
...             { case: { $gte: ["$marks", 70] }, then: "C" }
...           ],
...           default: "D"
...         }
...       }
...     }
...   }
... ])
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 8,
  modifiedCount: 8,
  upsertedCount: 0
}
```